EBU5376 Digital Signal Processing

LAB 1

Before you start, please read the following information carefully:

General Instructions:

- The .zip file you downloaded from QMPlus should contain the following documents:
  - This lab sheet (needs to be completed)
  - A WAV audio file that you need for both Question 1 and Question 2
- Type your answers in the textboxes provided. Figures need to be copied into the textboxes. If you need more space, enlarge the textbox or add a new box.
- Answers outside the textboxes will NOT be marked.
- If you require any technical assistance, please attend the lab session and ask the on-site TA for help.

How to submit:

- Submit one .zip file to QMPlus. The .zip file should contain:
  1. This LAB SHEET converted to a PDF document
  2. The MATLAB script containing all the work you did

Additional grading criteria:

- Your answers in this sheet must include sufficient explanations/derivations. Marks will be deducted if no explanation is given to support the answer.
- Your MATLAB script will be checked from the following three aspects:
  - Whether it can **REPRODUCE ALL THE FIGURES** in this sheet
  - Whether you have added appropriate **COMMENTS** to explain the codes
  - Plagiarism
- If the MATLAB script is not submitted, your grade will be capped to 60%.
- Email submission NOT accepted.
- Do NOT plagiarize. Both the MATLAB script and this sheet will be checked for plagiarism.

Question 1 – Basic Signal Processing **[25 marks]**

In this question, you will practice how to load audio files, process and visualize audio data, and explore the characteristics of sound using MATLAB.

In Lab 1 materials, you are given a WAV audio file that consists of a sequence of pure music notes.

Task 1: Use the command 'audioread( )' to read the audio in MATLAB, then answer the following questions:

1. What is the sampling rate of this signal?
2. How many samples are there in this signal?
3. What is the length (in seconds) of this signal?

(Optional: You can use the command 'soundsc( )' to play the audio in MATLAB.)

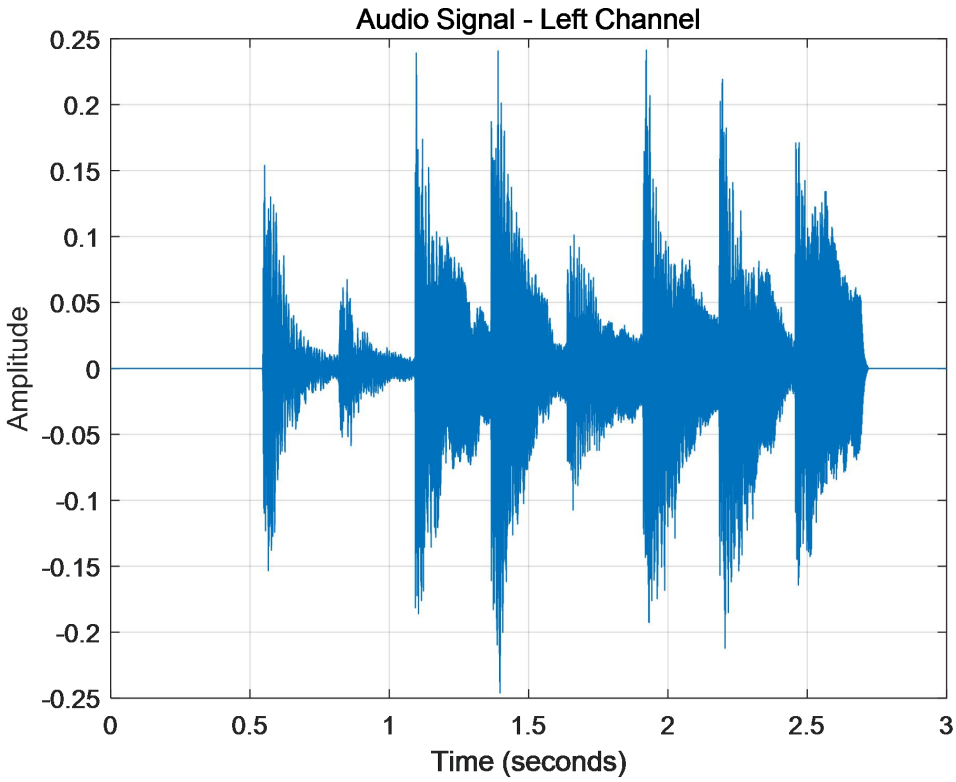
[4 marks]

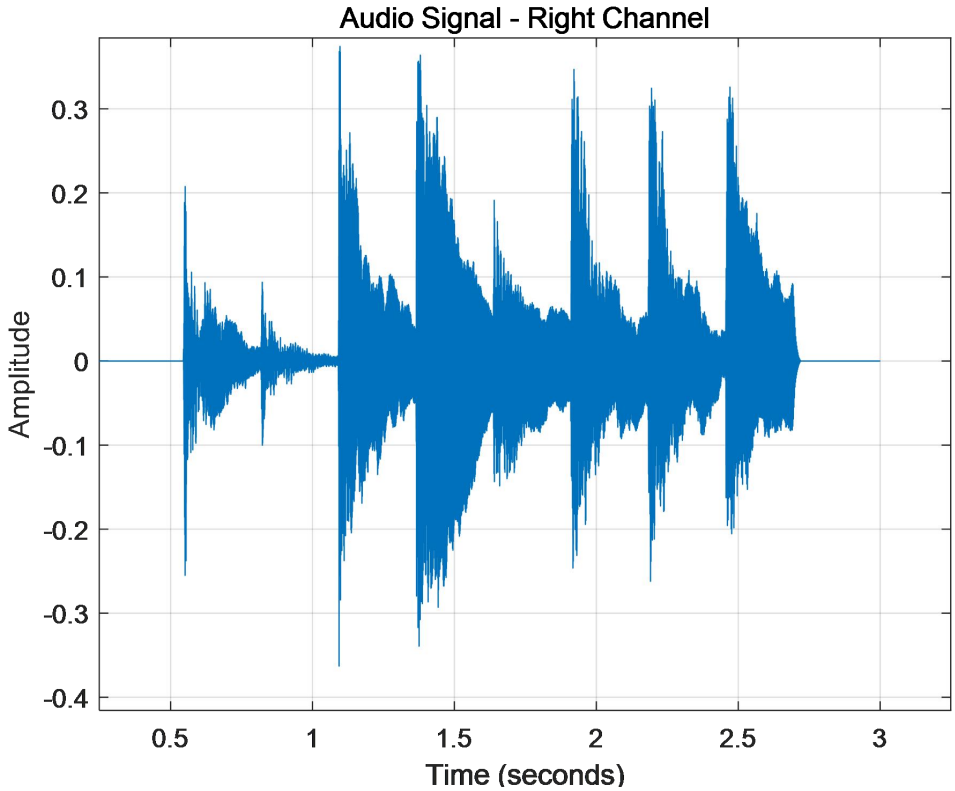
```
>> Lab1Q1

---------------------------------------------------

Q1 Task 1 Results:

---------------------------------------------------

1. The sampling rate is: 48000 Hz

2. The number of samples is: 144000

3. The length of the signal is: 3.00 seconds

---------------------------------------------------

---------------------------------------------------
```

Task 2: The provided audio file is stereo, with two channels (left channel and right channel) of audio data. Hence, the audio data should contain two dimensions that represent the samples over time and the stereo channels. On **two separate figures**, plot the left channel and the right channel of this audio signal, with the following requirements:

1. The horizontal axis is labelled in time (seconds).
2. Using titles or legends, clearly label whether the plot is the left channel or right channel.

[6 marks]





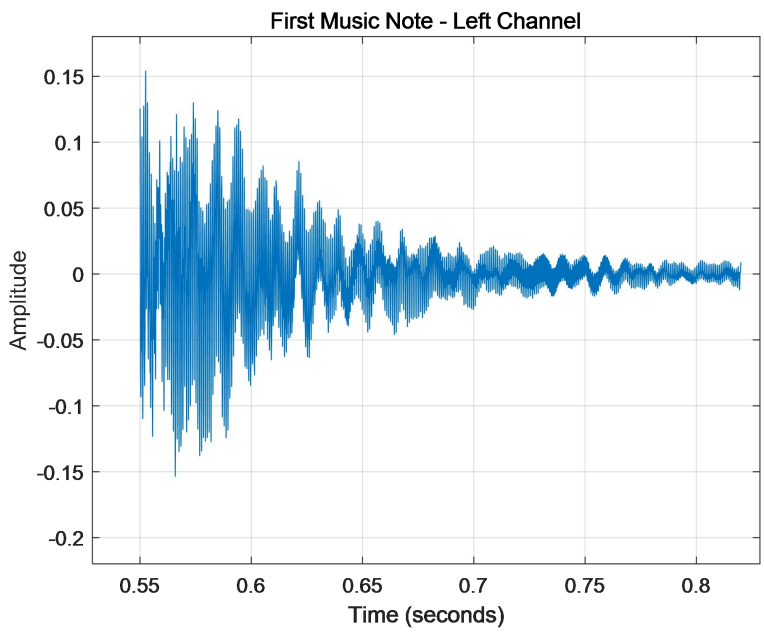
Task 3: The audio signal consists of several pure music notes. By inspecting the plot, how many music notes are there in this audio?

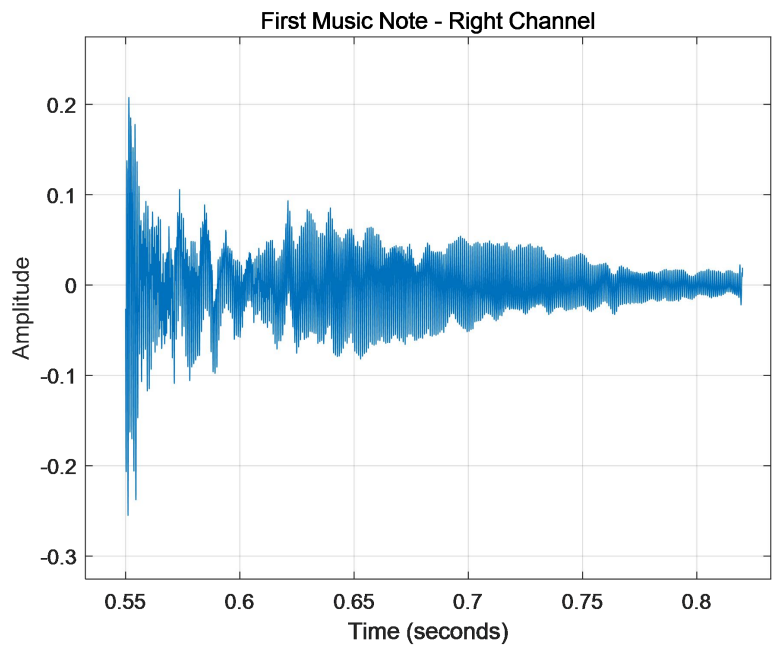
[1 mark]

Eight

Task 4: We would like to take a closer look at the **first music note**. On separate figures, plot the left and right channels for the segment where the first note is played. Clearly indicate whether the plot shows the left or right channel.

[6 marks]



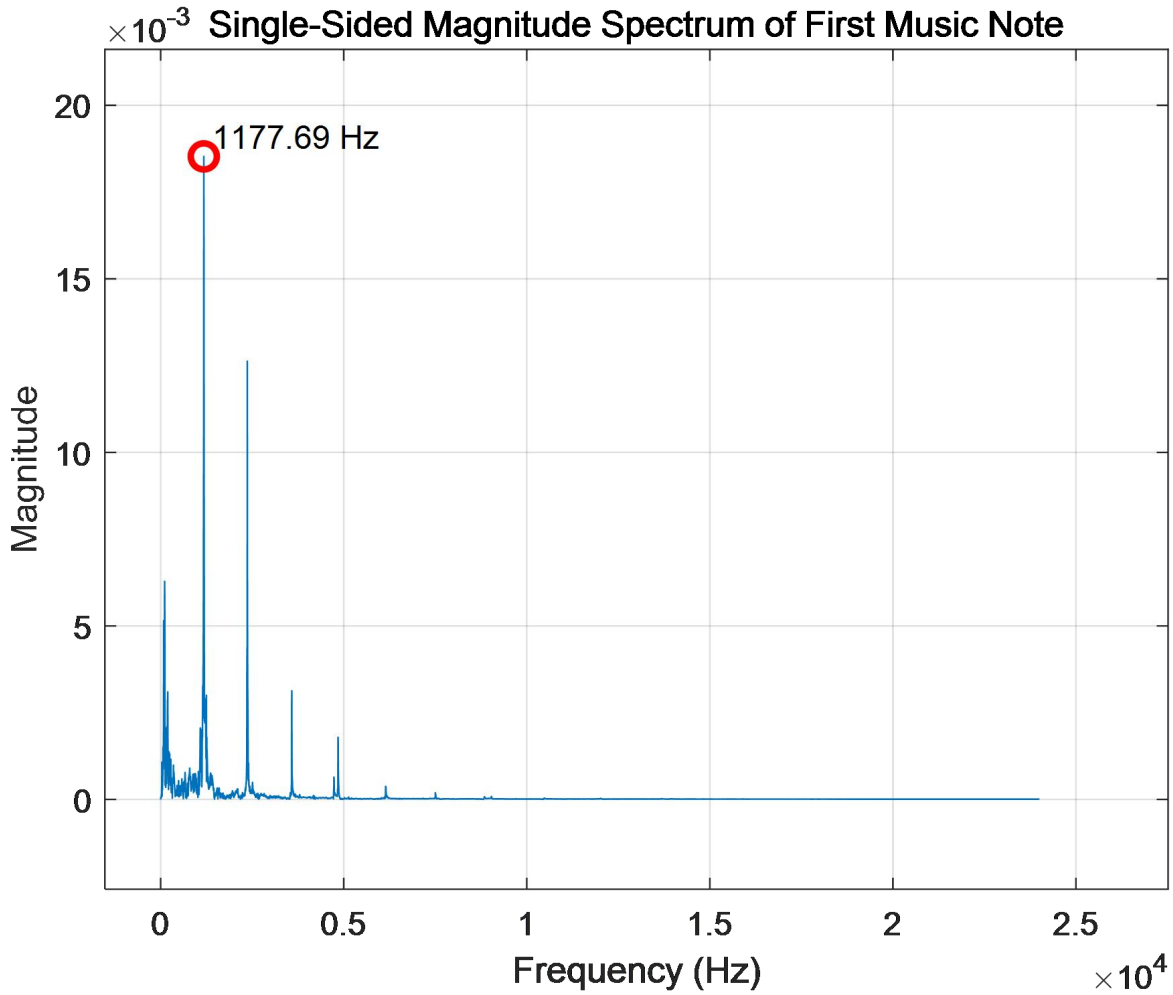


Task 5: Using the Fourier transform, estimate the frequency (in Hz) of the **first music note** in the given audio signal. Carefully explain the steps taken to obtain the answer, and include figures where relevant.

[8 marks]

To estimate the fundamental frequency of the first music note, I used the following steps:
1. The segment corresponding to the first note was extracted from the audio signal based on its time boundaries (identified by inspecting the full waveform plot).
2. The Fast Fourier Transform (FFT) was applied to this extracted segment to analyze its frequency content.
3. The single-sided magnitude spectrum was calculated from the FFT output. This spectrum shows the amplitude of different frequencies present in the note. The frequency axis was generated based on the sampling rate and the length of the segment.
4. The fundamental frequency was determined by finding the frequency corresponding to the highest peak in the magnitude spectrum (excluding the DC component at 0 Hz). This represents the most dominant frequency component, characteristic of a pure musical note.

Q1 Task 5 Result:
Estimated fundamental frequency of the first note: 1177.6869 Hz



Question 2 – Advanced Frequency Analysis **[25 marks]**

In this question, you will extend your analysis from Question 1 to estimate the frequencies of all music notes in the audio signal. You will also reflect on the process and discuss key observations and factors affecting the accuracy of your results.

Task 1: Use Fourier transform to estimate the frequency of all music notes in the given audio. Directly write down your results, you don't need to show your work here.

[8 marks]

Q2 Task 1 Results: Estimating Frequencies for Each Note Segment

---

Note 1 (0.5500 - 0.8200 sec, L=12960): Estimated Frequency = 1177.7778 Hz

Note 2 (0.8300 - 1.0800 sec, L=12000): Estimated Frequency = 1052.0000 Hz

Note 3 (1.0900 - 1.3600 sec, L=12960): Estimated Frequency = 833.3333 Hz

Note 4 (1.3600 - 1.6200 sec, L=12480): Estimated Frequency = 788.4615 Hz

Note 5 (1.6300 - 1.9000 sec, L=12960): Estimated Frequency = 700.0000 Hz

Note 6 (1.9100 - 2.1100 sec, L=9600): Estimated Frequency = 625.0000 Hz

Note 7 (2.1200 - 2.4300 sec, L=14880): Estimated Frequency = 590.3226 Hz

Note 8 (2.4500 - 2.7000 sec, L=12000): Estimated Frequency = 528.0000 Hz

---

Summary of Estimated Frequencies for Q2 Task 1:

1177.7778, 1052.0000, 833.3333, 788.4615, 700.0000, 625.0000, 590.3226, 528.0000 Hz

---

Task 2: Briefly describe how the **time window of the notes** (i.e., **start and end time of each note**) were decided in your answers.

[2 marks]

The time windows (start and end times) for each music note were decided by visually inspecting the full audio waveform plot (generated in Q1 Task 2 or Task 3). The plot displays the amplitude of the audio signal over time. Each music note corresponds to a segment with relatively high signal amplitude, which visually stands out from the surrounding periods of low amplitude (representing rests or silence). By observing these distinct, high-amplitude segments in the waveform, I identified the approximate start and end times on the horizontal (time) axis for each note and used these values to define the time intervals for subsequent analysis.

Note 1 (0.5500 - 0.8200 sec, L=12960): Estimated Frequency = 1177.7778 Hz

Note 2 (0.8300 - 1.0800 sec, L=12000): Estimated Frequency = 1052.0000 Hz

Note 3 (1.0900 - 1.3600 sec, L=12960): Estimated Frequency = 833.3333 Hz

Note 4 (1.3600 - 1.6200 sec, L=12480): Estimated Frequency = 788.4615 Hz

Note 5 (1.6300 - 1.9000 sec, L=12960): Estimated Frequency = 700.0000 Hz

Note 6 (1.9100 - 2.1100 sec, L=9600): Estimated Frequency = 625.0000 Hz

Note 7 (2.1200 - 2.4300 sec, L=14880): Estimated Frequency = 590.3226 Hz

Note 8 (2.4500 - 2.7000 sec, L=12000): Estimated Frequency = 528.0000 Hz

Task 3: For this particular audio signal, investigate how the duration of the time window used to isolate notes impacts the Fourier spectrum and the accuracy of estimated frequencies. Think about

1. How does the **peak frequency** change with window length? Is it stable or shifting?
2. What happens to the **frequency resolution** (width of peaks) in the spectrum as the window size changes?
3. How to set the window length for ensuring a reliable frequency estimate?

You need to show detailed analysis and figures to support your conclusion. Make sure your answers are clear and well-structured.

[15 marks]

To investigate the impact of window length on frequency analysis, I performed FFT analysis on the first music note segment (duration 0.2700 seconds, 12960 samples) using various window durations taken from the start of the note.

Answer to Q2 Task 3.1

The estimated fundamental frequency values showed some variation as the window duration changed. The estimated frequencies obtained for the tested durations (0.0100s, 0.0500s, 0.1000s, 0.1500s, and 0.2700s) were 1200.0000 Hz, 1180.0000 Hz, 1180.0000 Hz, 1180.0000 Hz, and 1177.7778 Hz, respectively. While most estimates were around 1180 Hz, the shortest window (0.0100s) gave a slightly higher estimate (1200 Hz), and the full note length window (0.2700s) gave a slightly lower estimate (1177.7778 Hz). This suggests that shorter windows might be less stable in estimating the exact peak frequency.

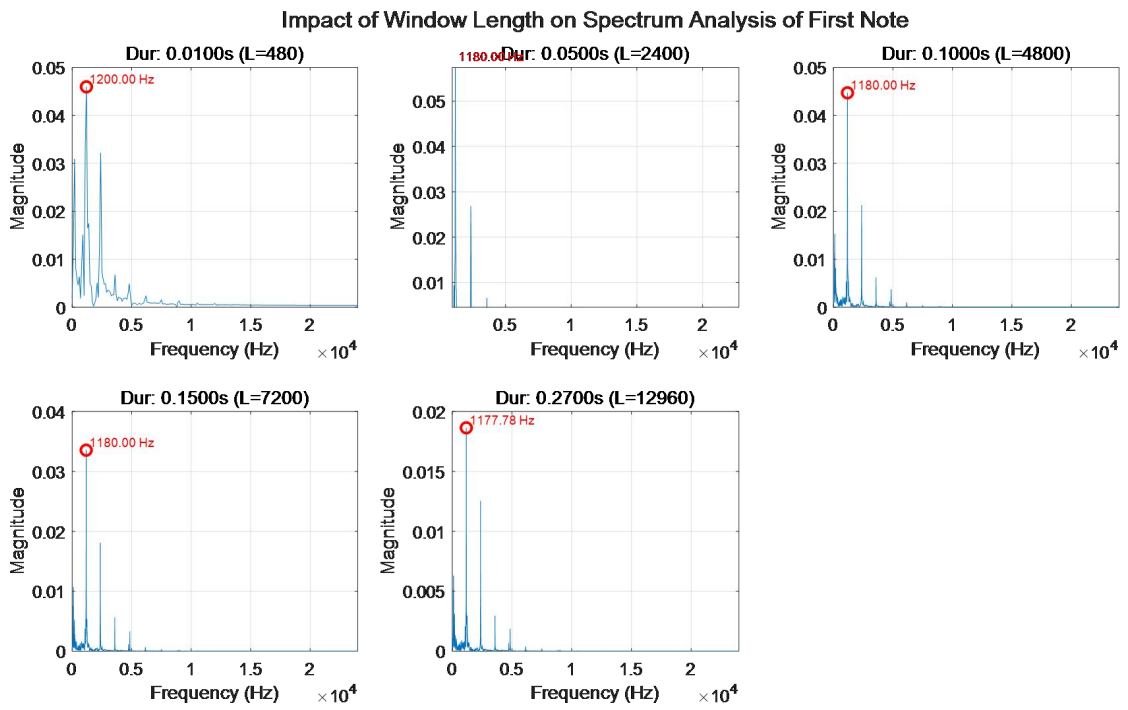
Answer to Q2 Task 3.2

The theoretical frequency resolution (fs/L) significantly improves (decreases numerically) as the window duration (and thus length L) increases. The resolutions for the tested durations were 100.0000 Hz (0.0100s), 20.0000 Hz (0.0500s), 10.0000 Hz (0.1000s), 6.6667 Hz (0.1500s), and 3.7037 Hz (0.2700s). This relationship is clearly demonstrated in the magnitude spectrum plots shown below. The shortest window (0.0100s) produces a very wide, broad peak in the spectrum. As the window duration increases, the peak becomes progressively narrower, taller, and more distinct, especially evident when comparing the 0.0100s plot to the 0.2700s plot.

Answer to Q2 Task 3.3

To ensure a reliable frequency estimate for a pure music note, the window length is critical. A very short window yields poor frequency resolution, resulting in a wide spectral peak that makes the precise fundamental frequency difficult to determine accurately and reliably. While longer windows provide better frequency resolution and sharper peaks, the window should ideally capture the note's stable portion without including significant transitions or silences. For this signal, using a window length that encompasses a substantial part, or the entirety of the stable note duration (like 0.2700s), provides the best frequency resolution and the sharpest peak, leading to a more reliable frequency estimation compared to very short windows.

EBU5376 24-25



Final Evaluation (optional): Copy and paste the following codes to MATLAB. Replace the first two lines with your results. Then, you will be able to hear the music sequence based on your estimation. You may compare it to the given audio.

```
% Final Evaluation (optional): Copy and paste the following codes to MATLAB.
% Replace f1-f8 with your estimations in Q2 task 1
frequencies = [1177.7778, 1052.0000, 833.3333, 788.4615, 700.0000, 625.0000, 590.3226, 528.0000];
% set this to the sampling rate of the audio, calculated in Q1 task 1
fs = 48000;
note_duration = 0.3;
rest_duration = 0.05;
fade_duration = 0.2;
audio_signal = [];
for i = 1:length(frequencies)
   % Generate the note waveform (sine wave)
   t = 0:1/fs:note_duration-1/fs;
   note = sin(2*pi*frequencies(i)*t);
   % Apply a fade-out to the end of the note to prevent clicking
   fade_samples = round(fade_duration*fs);
   % Ensure fade_samples is not longer than the note itself
   fade_samples = min(fade_samples, length(note));
   if fade_samples > 0
      % Create a fade-out window (linear ramp from 1 to 0) for the end
      fade_window_end = linspace(1, 0, fade_samples);
      % Apply the fade-out window to the end of the note
      note(end-fade_samples+1:end) = note(end-fade_samples+1:end) .* fade_window_end;
   end
   % Append the note to the overall audio signal
   audio_signal = [audio_signal, note];
   % Add a rest (silence) after the note, unless it's the last note
   if i < length(frequencies)
       rest_samples = round(rest_duration*fs);
       audio_signal = [audio_signal, zeros(1, rest_samples)];
   end
end
% Normalize the entire synthesized audio signal to prevent clipping during playback
audio_signal = audio_signal / max(abs(audio_signal));
% Play the sequence
```